

MATH 2123 (MATHEMATICS-III)

2016 Beginner Guide + Full Step-by-Step Solutions

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NOTE: This guide is written for a student who is not yet comfortable with the topics. Each set follows Topic -> Basic -> Problem-Solving Procedure -> Worked Solution. The equations and diagrams were read from the supplied three-page exam scan; where a symbol could be confused, the solution also checks the mathematics for consistency.

How to study from this PDF

- First read BASIC. Do not begin the calculation until the symbols and formula make sense.
- Memorize the PROCEDURE. The numerical data may change in another year, but the procedure usually stays the same.
- Then work through the solution with a pen. Cover the next line and try it yourself before reading it.
- Use the Rapid Revision pages at the end on the night before the exam.

Question map

Set	Main topic	Core skill
1	Matrix basics, Hermitian matrices, matrix polynomial	Definitions, conjugate transpose, powers of a matrix
2	Inverse, elementary transformations, rank/normal form	Gauss-Jordan elimination and row reduction
3	Parametric linear systems and homogeneous systems	Determinant/rank tests and null space
4	Network flow, eigen-system, symmetric inverse	Conservation equations and eigenvectors
5	Vector identities, parallelogram, dependence	Vector triple product and determinant/relation
6	Tangent plane, directional derivative, radial vector calculus	Gradient and radial derivatives
7	Projectile motion, line integral, triple integral	Integrate vectors and set correct limits
8	Laplace transform, inverse Laplace, forced ODE	Transform rules, partial fractions, convolution

Question 1: Matrix definitions, Hermitian decomposition, and a matrix polynomial

TOPIC

Matrix theory

BASIC

A matrix is a rectangular array. A scalar matrix is kl . Hermitian means $A^* = A$, where $*$ is conjugate transpose. Trace is the sum of diagonal entries.

PROCEDURE

For definitions: state the test and give a tiny example. For a matrix polynomial: compute A^2 and A^3 carefully, then combine corresponding entries.

1(a) - Definitions with examples

1) Matrix: a rectangular arrangement of numbers or functions in rows and columns.

Example: $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ is a 2×3 matrix.

2) Scalar matrix: a diagonal square matrix whose diagonal entries are all equal.

Example: $3I_3 = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$.

3) Hermitian matrix: a complex square matrix satisfying $A^* = A$, where A^* is the conjugate transpose.

Example: $\begin{bmatrix} 2 & i \\ -i & 3 \end{bmatrix}$ is Hermitian.

4) Trace: the sum of the main diagonal entries of a square matrix.

If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, then $\text{tr}(A) = 1 + 4 = 5$.

FINAL ANSWER: Know the four tests: matrix = rectangular array; scalar = kl ; Hermitian = $A^* = A$; trace = sum of diagonal elements.

1(b) - Express X as a Hermitian part plus a skew-Hermitian part

For any complex square matrix X , add and subtract its conjugate transpose X^* .

$$X = (1/2)(X + X^*) + (1/2)(X - X^*)$$

Define

$$H = (X + X^*)/2, \quad K = (X - X^*)/2$$

Check the two parts:

$$H^* = (X^* + X)/2 = H, \text{ so } H \text{ is Hermitian.}$$

$$K^* = (X^* - X)/2 = -K, \text{ so } K \text{ is skew-Hermitian.}$$

FINAL ANSWER: $X = H + K$, where $H = (X + X^*)/2$ and $K = (X - X^*)/2$.

1(c) - Evaluate $A^3 + 3A^2 - 7A + 9I$

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 3 & -1 \\ -3 & 1 & 2 \end{bmatrix}$$

First multiply A by itself:

$$A^2 = \begin{bmatrix} -8 & 3 & 9 \\ 11 & 8 & 1 \\ -7 & 5 & -6 \end{bmatrix}$$

Then multiply A^2 by A :

$$A^3 = \begin{bmatrix} -29 & 18 & -9 \\ 24 & 25 & 27 \\ 21 & 9 & -38 \end{bmatrix}$$

Now combine the four matrices entry by entry:

$$\begin{aligned} A^3 + 3A^2 - 7A + 9I \\ = \begin{bmatrix} -51 & 27 & -3 \\ 43 & 37 & 37 \\ 21 & 17 & -61 \end{bmatrix} \end{aligned}$$

FINAL ANSWER: $\begin{bmatrix} -51 & 27 & -3 \\ 43 & 37 & 37 \\ 21 & 17 & -61 \end{bmatrix}$.

Question 2: Inverse matrix, elementary transformations, and rank

TOPIC

Linear algebra computation

BASIC

A^{-1} exists only for a nonsingular square matrix ($\det A \neq 0$). Row operations: swap rows, multiply a row by nonzero k , add a multiple of one row to another. Rank is the number of pivots.

PROCEDURE

Inverse: augment $[A|I]$ and row-reduce to $[I|A^{-1}]$. System: then $x = A^{-1}b$. Rank/normal form: reduce to RREF, count pivots, and use column operations to reach $\text{diag}(I_r, 0)$.

2(a) - Find the inverse and solve the system

$$x - 3y - 8z = -10$$

$$3x + y = -4$$

$$2x + 5y + 6z = 13$$

$$A = \begin{bmatrix} 1 & -3 & -8 \\ 3 & 1 & 0 \\ 2 & 5 & 6 \end{bmatrix}, \quad b = \begin{bmatrix} -10 \\ -4 \\ 13 \end{bmatrix}^T$$

Start Gauss-Jordan on $[A|I]$. Useful operations are:

1. $R_2 \rightarrow R_2 - 3R_1$ and $R_3 \rightarrow R_3 - 2R_1$.
2. $R_3 \rightarrow 10R_3 - 11R_2$. This creates $[0, 0, -44]$ in row 3.
3. Scale R_3 by $-1/44$.
4. Use the new pivot in column 3 to clear column 3 above it.
5. Scale row 2 and clear column 2 above it.

$$A^{-1} = \begin{bmatrix} -3/22 & 1/2 & -2/11 \\ 9/22 & -1/2 & 6/11 \\ -13/44 & 1/4 & -5/22 \end{bmatrix}$$

Now multiply $A^{-1}b$:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix}^T = A^{-1}b = \begin{bmatrix} -3 \\ 5 \\ -1 \end{bmatrix}^T$$

FINAL ANSWER: A^{-1} as above; solution $x = -3$, $y = 5$, $z = -1$.

2(b) - Elementary transformations valid on a matrix

- Interchange two rows (or two columns): $R_i \leftrightarrow R_j$.
- Multiply one row (or column) by a nonzero scalar: $R_i \rightarrow kR_i$, $k \neq 0$.
- Add a multiple of one row (or column) to another: $R_i \rightarrow R_i + kR_j$.
- These operations preserve rank. Row operations also preserve the solution set of a linear system when applied to the augmented matrix.

FINAL ANSWER: The three basic types are interchange, nonzero scaling, and replacement by row/column plus a multiple of another.

2(c) - Row-reduced echelon form, normal form, and rank

$$A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

A convenient echelon form is

$$\begin{bmatrix} 2 & 3 & -1 & -1 \\ 0 & -5 & -3 & -7 \\ 0 & 0 & -66 & -44 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

There are three nonzero rows, so $\text{rank}(A)=3$. Continuing to reduced row-echelon form gives

$$\text{RREF}(A) = \begin{bmatrix} 1 & 0 & 0 & -5/3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2/3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

To reach normal form, remove the last column using a column operation:

$$C_4 \rightarrow C_4 + (5/3)C_1 - C_2 - (2/3)C_3$$

$$\text{Normal form} = \text{diag}(1, 1, 1, 0)$$

FINAL ANSWER: $\text{rank}(A)=3$; normal form is $\text{diag}(I_3, 0)$.

Question 3: Systems depending on a parameter and homogeneous systems

TOPIC

Consistency and null space

BASIC

For $Ax=b$: unique if $\det A \neq 0$ (square case). If $\det A=0$, compare ranks. A homogeneous system $Ax=0$ is always consistent because $x=0$ always works.

PROCEDURE

Parameter problem: compute $\det A$ as a function of λ . At special roots, row-reduce $[A|b]$. Homogeneous problem: row-reduce A , choose free variables, and express the general solution as a linear combination of basis vectors.

3(a) - Values of λ for unique and infinitely many solutions

$$3x - y + 4z = 3$$

$$x + 2y - 3z = -2$$

$$6x + 5y + \lambda z = -3$$

Coefficient determinant:

$$\det A = 7(\lambda + 5)$$

Therefore, if $\lambda \neq -5$, $\det A \neq 0$ and the solution is unique.

At $\lambda = -5$, row reduction gives

$$\text{RREF}([A|b]) = \begin{bmatrix} 1 & 0 & 5/7 & 4/7 \\ 0 & 1 & -13/7 & -9/7 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Let $z=t$. Then

$$x = 4/7 - (5/7)t, \quad y = -9/7 + (13/7)t, \quad z = t$$

FINAL ANSWER: Unique for $\lambda \neq -5$. Infinitely many for $\lambda = -5$, with $(x, y, z) = (4/7 - 5t/7, -9/7 + 13t/7, t)$.

3(b) - Why homogeneous systems are always consistent; solve the given system

A homogeneous system has the form $Ax=0$. The zero vector $x=0$ always satisfies it, so it is always consistent. The zero solution is called the trivial solution; any solution other than zero is non-trivial.

$$4x_1 - x_2 + 2x_3 + x_4 = 0$$

$$2x_1 + 3x_2 - x_3 - 2x_4 = 0$$

$$7x_2 - 4x_3 - 5x_4 = 0$$

$$2x_1 - 11x_2 + 7x_3 + 8x_4 = 0$$

The reduced matrix is

$$[[1, 0, 5/14, 1/14], [0, 1, -4/7, -5/7], [0, 0, 0, 0], [0, 0, 0, 0]]$$

Take $x_3=s$ and $x_4=t$. Then

$$x_1 = -(5/14)s - (1/14)t, \quad x_2 = (4/7)s + (5/7)t$$

$$(x_1, x_2, x_3, x_4) = s(-5/14, 4/7, 1, 0) + t(-1/14, 5/7, 0, 1)$$

FINAL ANSWER: Two free parameters; equivalently $\text{span}\{(-5, 8, 14, 0), (-1, 10, 0, 14)\}$.

3(c) - Test the given matrix for orthogonality

$$Q = (1/3)[[2, 2, 1], [-2, 1, 2], [1, -2, 2]]$$

A real square matrix is orthogonal if $Q^T Q = I$. The unscaled rows have length 3 and are mutually perpendicular:

$$[[2, 2, 1], [-2, 1, 2], [1, -2, 2]]^T \cdot \text{same} = 9I$$

$$\text{Therefore } Q^T Q = (1/9)(9I) = I.$$

FINAL ANSWER: Yes. The matrix is orthogonal.

Question 4: Network flow, eigenvalues/eigenvectors, and a symmetry proof

TOPIC	BASIC	PROCEDURE
Conservation + eigensystem	At every network node: inflow = outflow. Eigenvalues solve $\det(A - \lambda I) = 0$; eigenvectors solve $(A - \lambda I)v = 0$. For matrices, $(A^{-1})^T = (A^T)^{-1}$.	Network: write one node balance at a time and solve in terms of any free flow. Eigen: characteristic polynomial first, then one null-space calculation per eigenvalue. Symmetry proof: transpose the identity $AA^{-1} = I$.

4(a) - Set up the network equations and identify the solution type

Reading the arrow directions in the supplied diagram gives these node balances:

$$\text{Node 1: } 20 = x_1 + x_2$$

$$\text{Node 2: } x_4 = 20 + x_3$$

$$\text{Node 3: } x_2 + x_3 = 20$$

$$\text{Node 4: } x_1 + 10 = x_5$$

$$\text{Node 5: } x_5 + 10 = x_4$$

Let $x_2=t$. Then the other flows are

$$x_1=20-t, \quad x_3=20-t, \quad x_5=30-t, \quad x_4=40-t$$

The fifth equation is automatically satisfied, so one variable is free. If physical traffic flows must be nonnegative, $0 \leq t \leq 20$.

FINAL ANSWER: Infinitely many solutions: $(x_1, x_2, x_3, x_4, x_5) = (20-t, t, 20-t, 40-t, 30-t)$, normally with $0 \leq t \leq 20$.

4(b) - Eigenvalues and corresponding eigenvectors

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$$

Characteristic polynomial:

$$\det(A - \lambda I) = (\lambda - 6)(\lambda - 3)(\lambda + 2)$$

Hence $\lambda = 6, 3, -2$. Solving $(A - \lambda I)v = 0$ gives convenient eigenvectors:

$$\lambda = 6: v = (1, 2, 1)$$

$$\lambda = 3: v = (1, -1, 1)$$

$$\lambda = -2: v = (-1, 0, 1)$$

FINAL ANSWER: Eigenpairs: $(6, (1, 2, 1))$, $(3, (1, -1, 1))$, $(-2, (-1, 0, 1))$, up to nonzero scalar multiples.

4(c) - If A is symmetric and nonsingular, prove A^{-1} is symmetric

Since A is symmetric, $A^T = A$. For every nonsingular matrix,

$$(A^{-1})^T = (A^T)^{-1}$$

Substitute $A^T = A$:

$$(A^{-1})^T = A^{-1}$$

FINAL ANSWER: A^{-1} is symmetric.

Question 5: Vector triple product, rhombus from diagonals, and linear dependence

TOPIC

Vector algebra

BASIC

The vector triple product is $A \times (B \times C) = B(A \cdot C) - C(A \cdot B)$. A parallelogram is a rhombus when its diagonals are perpendicular. Linear dependence means a nonzero combination equals zero.

PROCEDURE

Use the triple-product identity algebraically. For the rhombus, dot the diagonals and then form side vectors $(A+B)/2$ and $(A-B)/2$. For dependence, look for a relation or compute a determinant.

5(a) - Prove the vector triple product identity and Jacobi identity

The standard vector triple product identity is

$$A \times (B \times C) = B(A \cdot C) - C(A \cdot B)$$

Apply it cyclically:

$$A \times (B \times C) = B(A \cdot C) - C(A \cdot B)$$

$$B \times (C \times A) = C(A \cdot B) - A(B \cdot C)$$

$$C \times (A \times B) = A(B \cdot C) - B(A \cdot C)$$

Adding the three equations cancels every term.

FINAL ANSWER: $A \times (B \times C) + B \times (C \times A) + C \times (A \times B) = 0$.

5(b) - Show the parallelogram is a rhombus and find its side length

Diagonal vectors: $A = (3, -4, -1)$, $B = (2, 3, -6)$

Their dot product is

$$A \cdot B = 3(2) + (-4)(3) + (-1)(-6) = 6 - 12 + 6 = 0$$

So the diagonals are perpendicular. A parallelogram with perpendicular diagonals is a rhombus.

The two adjacent side vectors are half the sum and half the difference of the diagonals:

$$u = (A+B)/2 = (5, -1, -7)/2$$

$$v = (A-B)/2 = (1, -7, 5)/2$$

$$|u| = |v| = (1/2)\sqrt{75} = 5\sqrt{3}/2$$

FINAL ANSWER: It is a rhombus; each side has length $5\sqrt{3}/2$.

5(c) - Determine linear dependence

$A = (2, 1, -3)$, $B = (1, 0, -4)$, $C = (4, 3, -1)$

A direct relation is

$$-3A + 2B + C = 0$$

Since the coefficients are not all zero, the three vectors are linearly dependent.

FINAL ANSWER: Dependent; $C = 3A - 2B$.

Question 6: Tangent plane, maximum directional derivative, and radial formulas

TOPIC

Gradient geometry

BASIC

For a level surface $F=0$, $\text{grad } F$ is normal to the tangent plane. The maximum directional derivative of ϕ is $|\text{grad } \phi|$ and occurs in the $\text{grad } \phi$ direction. For $r=\sqrt{x^2+y^2+z^2}$, $\text{grad } f(r)=f'(r) \mathbf{r_vector}/r$.

PROCEDURE

Tangent plane: define F , compute grad , evaluate at point. Directional derivative: compute $\text{grad } \phi$, then normalize it. Radial problems: use the 3D radial Laplacian $f''+2f/r$.

6(a) - Tangent plane and normal line to $z=x^2+y^2$ at $(2,-1,5)$

$$F(x,y,z)=x^2+y^2-z$$

$$\text{grad } F=(2x,2y,-1)$$

$$\text{grad } F(2,-1,5)=(4,-2,-1)$$

Tangent plane:

$$4(x-2)-2(y+1)-(z-5)=0$$

$$4x-2y-z=5$$

Normal line through the point in direction $(4,-2,-1)$:

$$(x-2)/4 = (y+1)/(-2) = (z-5)/(-1)$$

FINAL ANSWER: Plane: $4x-2y-z=5$. Normal line: $(x-2)/4=(y+1)/(-2)=(z-5)/(-1)$.

6(b) - Direction and magnitude of maximum directional derivative

$$\phi=x^2 y z^3$$

$$\text{grad } \phi=(2xyz^3, x^2z^3, 3x^2yz^2)$$

$$\text{grad } \phi(2,1,-1)=(-4,-4,12)=4(-1,-1,3)$$

$$|\text{grad } \phi|=\sqrt{16+16+144}=4\sqrt{11}$$

$$\text{Unit direction of maximum increase} = (-1,-1,3)/\sqrt{11}$$

FINAL ANSWER: Direction = $(-i-j+3k)/\sqrt{11}$; maximum directional derivative = $4\sqrt{11}$.

6(c) - Evaluate (i) Laplacian of $\ln r$ and (ii) $\text{grad } f(r)$

For a radial scalar $f(r)$ in three dimensions,

$$\nabla^2 f(r) = f''(r) + (2/r)f'(r), \quad r \neq 0$$

Take $f(r)=\ln r$: $f'=1/r$ and $f''=-1/r^2$.

$$\nabla^2(\ln r) = -1/r^2 + 2/r^2 = 1/r^2$$

$$\text{grad } f(r) = f'(r) \text{grad } r = f'(r)(x \mathbf{i} + y \mathbf{j} + z \mathbf{k})/r$$

FINAL ANSWER: (i) $\nabla^2(\ln r)=1/r^2$. (ii) $\text{grad } f(r)=f'(r) \mathbf{r_vector}/r$, for $r \neq 0$.

Question 7: Projectile motion, line integrals, and a triple integral

TOPIC

Vector integration and multiple integrals

BASIC

Acceleration integrates to velocity; velocity integrates to position. A line integral depends on the path unless the field is conservative. Triple integrals require limits that describe the solid.

PROCEDURE

Projectile: integrate componentwise and apply initial data. Line integral: parameterize each segment or the direct line. Triple integral: sketch inequalities, choose an order, then integrate from inside outward.

7(a) - Velocity and displacement under constant gravitational acceleration

$$\mathbf{a} = -g \mathbf{j}$$

Integrate once:

$$\mathbf{v}(t) = C_1 \mathbf{i} + (C_2 - gt) \mathbf{j}$$

At $t=0$, $\mathbf{v}(0) = V_0 \cos(\theta_0) \mathbf{i} + V_0 \sin(\theta_0) \mathbf{j}$, so

$$\mathbf{v}(t) = V_0 \cos(\theta_0) \mathbf{i} + [V_0 \sin(\theta_0) - gt] \mathbf{j}$$

Integrate again and use $\mathbf{r}(0) = \mathbf{0}$:

$$\mathbf{r}(t) = V_0 \cos(\theta_0) t \mathbf{i} + [V_0 \sin(\theta_0) t - \frac{1}{2} g t^2] \mathbf{j}$$

FINAL ANSWER: $\mathbf{v}(t) = V_0 \cos(\theta_0) \mathbf{i} + (V_0 \sin(\theta_0) - gt) \mathbf{j}$; $\mathbf{r}(t) = V_0 \cos(\theta_0) t \mathbf{i} + (V_0 \sin(\theta_0) t - \frac{1}{2} g t^2) \mathbf{j}$.

7(b) - Line integral along two paths

$$\mathbf{A} = (2y+3) \mathbf{i} + xz \mathbf{j} + (yz-x) \mathbf{k}$$

Path (i): three axis-aligned straight segments.

$$C_1: (0,0,0) \rightarrow (0,0,1): \text{integral} = 0$$

$$C_2: (0,0,1) \rightarrow (0,1,1): \text{integral} = 0$$

$$C_3: (0,1,1) \rightarrow (2,1,1): \text{integral} = \int_0^2 5 \, dx = 10$$

Therefore integral along path (i) = 10.

Path (ii): direct line from $(0,0,0)$ to $(2,1,1)$. Parameterize $\mathbf{r}(t) = (2t, t, t)$, $0 \leq t \leq 1$.

$$d\mathbf{r} = (2, 1, 1) dt$$

$$\mathbf{A}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = 3t^2 + 2t + 6$$

$$\int_0^1 (3t^2 + 2t + 6) dt = 1 + 1 + 6 = 8$$

FINAL ANSWER: Path (i): 10. Path (ii): 8. Different values show the field is path-dependent.

7(c) - Evaluate the triple integral

Integral over V of $(2x+y) dV$, where $z = 4 - x^2$, $x=0$, $y=0$, $y=2$, $z=0$

Since $z \geq 0$ and $x \geq 0$, $4 - x^2 \geq 0$ gives $0 \leq x \leq 2$. Thus

$$0 \leq x \leq 2, \quad 0 \leq y \leq 2, \quad 0 \leq z \leq 4 - x^2$$

$$\begin{aligned} I &= \int_0^2 \int_0^2 \int_0^{4-x^2} (2x+y) \, dz \, dy \, dx \\ &= \int_0^2 (4x+2)(4-x^2) \, dx \\ &= 80/3 \end{aligned}$$

FINAL ANSWER: 80/3.

Question 8: Laplace transforms, inverse Laplace, and a forced differential equation

TOPIC	BASIC	PROCEDURE
Laplace transform	$L\{f'\} = sF - f(0)$, $L\{f''\} = s^2F - sf'(0) - f''(0)$. Standard transforms convert polynomials, exponentials and sines/cosines. A product $1/(s^2+a^2)$ with a forcing transform leads naturally to convolution.	Transform term-by-term. For inverse Laplace, use partial fractions. For the forced ODE, transform derivatives, isolate $Y(s)$, invert the initial-condition part, then write the forcing part as a convolution integral.

8(a) - Find the requested Laplace transforms

(i) If $L\{F(t)\} = \bar{F}(s)$, then

$$L\{F''(t)\} = s^2 \bar{F}(s) - sF(0) - F'(0)$$

(ii) For $e^{4t} + 4t^3 - 2\sin 3t + 4\cos 5t$:

$$L\{e^{4t}\} = 1/(s-4)$$

$$L\{4t^3\} = 4(3!)/s^4 = 24/s^4$$

$$L\{-2\sin 3t\} = -6/(s^2+9)$$

$$L\{4\cos 5t\} = 4s/(s^2+25)$$

FINAL ANSWER: $1/(s-4) + 24/s^4 - 6/(s^2+9) + 4s/(s^2+25)$, and the derivative rule above for F'' .

8(b) - Evaluate $L^{-1}\{1/[s^3(s^2+4)]\}$

Use partial fractions:

$$1/[s^3(s^2+4)] = -1/(16s) + 1/(4s^3) + s/[16(s^2+4)]$$

Invert each term:

$$L^{-1}\{1/s\} = 1, \quad L^{-1}\{1/s^3\} = t^2/2, \quad L^{-1}\{s/(s^2+4)\} = \cos 2t$$

$$f(t) = -1/16 + t^2/8 + (1/16)\cos 2t$$

FINAL ANSWER: $t^2/8 + (\cos 2t - 1)/16$.

8(c) - Solve $y'' + a^2 y = F(t)$, $y(0)=1$, $y'(0)=-2$

Let $Y(s)=L\{y(t)\}$ and $Fbar(s)=L\{F(t)\}$.

$$L\{y''\} = s^2 Y - s + 2$$

$$(s^2 + a^2)Y - s + 2 = Fbar(s)$$

$$Y = s/(s^2 + a^2) - 2/(s^2 + a^2) + Fbar(s)/(s^2 + a^2)$$

Invert the first two terms directly. For the last term, use convolution because

$$L^{-1}\{1/(s^2 + a^2)\} = (1/a)\sin(at)$$

$$y(t) = \cos(at) - (2/a)\sin(at) + (1/a) \int_0^t F(\tau) \sin[a(t-\tau)] d\tau$$

FINAL ANSWER: $y(t) = \cos(at) - (2/a)\sin(at) + (1/a) \int_0^t F(\tau) \sin[a(t-\tau)] d\tau$.

Rapid Revision: procedures to memorize

Topic	What to remember
Hermitian split	$H=(A+A^*)/2$ and $K=(A-A^*)/2$.
Inverse by row operations	$[A I] \rightarrow [I A^{-1}]$. Check $\det(A) \neq 0$ first.
Rank	Number of pivots/nonzero rows in echelon form.
Parameter system	Compute $\det$. At $\det=0$, compare ranks of A and $[A b]$.
Homogeneous system	Always consistent. Nontrivial solutions occur when there are free variables.
Orthogonal matrix	$Q^T Q = I$, so $Q^{-1} = Q^T$.
Network flow	At every node: inflow=outflow. One redundant equation often means one free flow.
Eigen-system	$\det(A - \lambda I) = 0$, then $(A - \lambda I)v = 0$.
Vector triple product	$A \times (B \times C) = B(A \cdot C) - C(A \cdot B)$.
Tangent plane	$F=0$; normal= $\text{grad } F$; plane $\text{grad } F(P) \cdot (r-P) = 0$.
Maximum directional derivative	Direction= $\text{grad } f / \text{grad } f $; maximum= $ \text{grad } f $.
Line integral	Parameterize path; integrate $A(r(t)) \cdot r'(t)$.
Triple integral	Translate all bounding surfaces into nested limits.
Laplace derivative	$L\{y''\} = s^2 Y - sy(0) - y'(0)$.
Convolution	$L^{-1}\{F(s)G(s)\} = \int_0^t f(\tau)g(t-\tau)d\tau$.

Answers at a glance

- 1(c) $[[-51, 27, -3], [43, 37, 37], [21, 17, -61]]$.
- 2(a) $(x, y, z) = (-3, 5, -1)$.
- 2(c) rank=3.
- 3(a) unique for $\lambda \neq -5$; infinite for $\lambda = -5$.
- 3(b) two-parameter family; basis can be $(-5, 8, 14, 0)$, $(-1, 10, 0, 14)$.
- 3(c) orthogonal.
- 4(a) infinite network flows: $(20-t, t, 20-t, 40-t, 30-t)$.
- 4(b) eigenvalues 6, 3, -2 with eigenvectors $(1, 2, 1)$, $(1, -1, 1)$, $(-1, 0, 1)$.
- 5(b) rhombus side length $5\sqrt{3}/2$.
- 5(c) dependent, $C=3A-2B$.
- 6(a) tangent plane $4x-2y-z=5$.
- 6(b) max direction $(-1, -1, 3)/\sqrt{11}$, magnitude $4\sqrt{11}$.
- 6(c) $\nabla^2 \ln r = 1/r^2$; $\text{grad } f(r) = f'(r)r_{\text{vector}}/r$.
- 7(a) standard projectile velocity and position formulas given above.
- 7(b) path integrals 10 and 8.
- 7(c) $80/3$.
- 8(b) $t^2/8 + (\cos 2t - 1)/16$.
- 8(c) $\cos(at) - (2/a)\sin(at) + (1/a) \int_0^t F(\tau)\sin(a(t-\tau))d\tau$.

Beginner exam strategy

- Do the definition/proof questions first if you remember the statements; they are faster than long row reductions.
- In a matrix problem, write the row operation beside every step. This protects method marks.
- For a parameter system, never guess from the equations: determinant first, rank second.
- For gradient questions, always evaluate the gradient at the point before normalizing.
- For line/triple integrals, spend one minute deciding the path or limits before integrating.
- For Laplace transforms, write the transform of each term separately; combine only after each term is correct.